

PROJECTIONS OF POINTS

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[GENERAL PROBLEMS]

[1] Draw the projections of the following points on the **same XY line** keeping **convenient distance between each projections**. Name the quadrants in which they lie.

- **A – 30 mm above HP and 35 mm in front of VP.**
- **B – 35 mm above HP and 40 mm behind VP.**
- **C – 40 mm above HP and on VP.**
- **D – 35 mm below HP and 30 mm in front of VP.**

[2] Draw the projections of the following points on the **same XY line** keeping **convenient distance between each projections**. Name the quadrants in which they lie.

- **E – 30 mm below HP and 25 mm behind VP.**
- **F – 35 mm below HP and 30 mm in front of VP.**
- **G – On HP and 30 mm in front of VP.**
- **H – On HP and 35 mm behind VP.**

[PROBLEMS ON 1ST QUADRANT]

[3] A point P is 30mm in front of VP, 40mm above HP and 50mm From RPP. Draw its projections and name the side view.

[4] A point P is 30 mm in front of VP, 20 mm above HP and 25 mm in front/behind/from LPP. Draw its projections and name the side view.

[5] Draw the projections of point G while is in 1st quadrant such that it is equidistant from both HP and VP. The point is 25mm from RPP. Determine its distance from HP and VP.

Note:

Left side view [LSV] → Right Profile Plane [RPP]

Right side view [RSV] → Left Profile Plane [LPP]

[PROBLEMS ON 3rd QUADRANT]

[6] A point is 35mm below HP, 20mm behind VP and 25mm behind/in front/from RPP.

Draw its projections and name the side view.

[7] A point is 40mm below HP, 20mm behind VP and 25mm behind/in front/from RPP.

Draw its projections and name the side view.

[8] A point G is 25mm below HP and is situated in the third quadrant is shortest distance from the intersection of XY and X_1Y_1 is 45mm. Draw its projection and find its distance from VP.

Note:

Left side view [LSV] → Left Profile Plane [LPP]

Right side view [RSV] → Right Profile Plane [RPP]

[PROBLEMS ON 2nd QUADRANT]

[9] A point P is 45mm above HP, 60mm behind VP and 30mm from RPP. Draw the three principle views of the point. Also, state the quadrant in which it lies.

[10] A point is 40mm behind VP, 20mm above HP and 30mm in front/behind/from LPP.
Draw its projections and name the side view.

Note:

Left side view [LSV] → Left Profile Plane [LPP]

Right side view [RSV] → Right Profile Plane [RPP]

[PROBLEMS ON 4th QUADRANT]

[11] Draw all the three views of a **point P** lying 60mm below HP, 70mm in front of VP and 40mm from RPP. Also, state the quadrant in which it lies.

[12] Draw all the three views of a **point P** lying 40mm below HP, 20mm in front of VP and 50mm from LPP. Also, state the quadrant in which it lies.

[13] A **point A** is 40mm in front of VP and is situated in the fourth quadrant. Its shortest distance from the intersection of XY and X_1Y_1 is 45mm. draw its projections. Also, find its distance from HP.

Note:

Left side view [LSV] → Right Profile Plane [RPP]

Right side view [RSV] → Left Profile Plane [LPP]

[SPECIAL PROBLEMS (more than one points)]

[14] A point **30mm above XY line** is the front view of two points A and B, the top view of A is 40mm behind VP and **the top view of B is 45mm in front of VP**. Draw the projections of the points and state the quadrants in which the points are situated.

[15] A point **30mm above XY line** is the front view of 3 points P, Q and R. the top view of R is 40mm behind VP, the **top view Q is on XY line** and **top view of point P is 45mm in front of VP**. Draw the projections of the points and state the quadrants in which the points are situated.

[16] A point **20mm below** the reference XY line is the top view of three points P, Q and R. P is 20mm below HP, Q is **35mm above HP** and **R is on HP**. Draw the projections of the three points and state their positions and quadrants in which they are situated.

[SPECIAL PROBLEMS (more than one points) (distance between their projectors = 0) (side view)]

[17] A point A is 20mm above HP and 25mm in front of VP. Another point B is 25mm behind VP and 40mm below HP. Draw, their projections when the distance between their projectors parallel to XY line is zero mm. add the right side view only to point B.

[18] The common point 40mm below XY line represents not only the front view of three points A, B and C but also the top view of point C. the top view of point B lies on XY line and top view of point A lies 50mm above it. Draw the projections of the points and add the right side view to the point A only. Also state in which the quadrant the points lie.

[SPECIAL PROBLEMS (more than one points) (distance between their projectors $\neq 0$)]

[19] A point **P** is on HP and 35mm in front of VP. Another point **Q** is on VP and below HP. The line joining their front views makes an angle of 30° to XY line, while the line joining their top views makes an angle of 45° with XY line. Find the distance between their projectors measured parallel to XY line. Also, find the distance of the point Q from HP.

[20] A point **R** is 25mm above HP and 20mm in front of VP. Another point **S** is on HP and 30mm behind VP. The distance between their projectors, measured parallel to the line of intersection VP and HP is 50mm. Find the distance between top and front views of point R and S.

[21] Two points **R** and **S** are on HP. The point R is 35mm in front of VP, while S is 50mm behind VP. The line joining their top views makes an angle of 40° with XY. Find the horizontal distance between the two projectors.

[SPECIAL PROBLEMS (more than one points) (distance between their projectors $\neq 0$) (side view)]

[22] A point M is 30mm in front of VP and 20mm above HP, another point N is 15mm behind VP and 25mm below HP. The horizontal distance between the points parallel to XY line is 50mm. Draw the projections of the points M and N and join their front and top views. Draw (i) The right side view for the point N.

(ii) The right side view for the point N and left side view for the point M.

[23] A point A is 30mm in front of VP and 40mm above HP. Another point B is 20mm behind VP and 35mm below HP. The horizontal distance between the points measured parallel to XY line is 60mm. draw the three projections of the points. Join their front and top views.

[PROBLEMS ON 1ST QUADRANT]

[24] A point S is in first quadrant and equidistant of 50mm from all the three principal planes. Draw the projections of the point. Draw all the three views of the point.